

Project 2, Geometric Transformation and Basic Scene Design

A car in a scene

1. Design a scene which includes the following items: (50%)
 - (a) A floor with ramps and flat areas, a car with a body and 4 wheels, and some simple buildings standing on the floor.
2. Basic functionalities of the car: (40%)
 - (a) Move forward and backward,
 - (b) Turn left and right,
 - (c) Avoid the buildings and other obstacles,
 - (d) Go uphill and downhill.
3. You can show the world coordinate system and the local coordinate system of the car in your scene such that the relative position of the car is revealed. (10%)
4. Extra credits: (10%)
 - (a) Any fancy ideas are welcome. You can earn 10 points at most by adding more functions or features into the car. (和同學相同的 fancy ideas 不計分)
5. Other requirements:
 - (a) Draw a simple scene graph of your scene in papers,
 - (b) Explain the procedure of controlling the car motions. List the functionalities of your car. Write down any extra efforts effects that you have implemented in your program to earn the extra credits.
 - (c) Hand in your programs and papers two weeks later. Demo your program in the graphics lab.
6. You may modify our sample programs for this homework. But you must design your own car and scene. If you hand in the sample programs, your will receive no credit at all.

(Assume the floor is located at the zx -plane. The vertical axis is the y -axis. The 2 opposite corner of the floor are at $(0, 0, 0)$ and $(100, 0, 100)$).

`/*In the display() callback func---*/`

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
gluLookAt(95.0, 70.0, 95.0, 45.0, 0.0, 4.0, 0.0, 1.0, 0.0);
```

`//In the reshape callback function`

```
void my_reshape(int w, int h)
{
    glViewport(0, 0, w, h);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    glOrtho(-50.0, 50.0, -50, 50, 0.0, 120.0);
}
```